AISC DG11 Floor Vibration Analysis Example 001

OFFICE FLOOR WITH CANTILEVER AREA – VIBRATIONS DUE TO WALKING

PROBLEM DESCRIPTION

The purpose of this example is to verify the Floor Vibration Analysis implemented in the program. The methodology adheres to the procedures outlined in Chapter 7—Finite Element Analysis Methods—of the AISC Steel Design Guide 11: Vibrations of Steel-Framed Structural Systems Due to Human Activity, Second Edition (referred to as the AISC DG11).

Example 7.1 presented in the AISC DG11 is used to verify the Floor Vibration Analysis Implementation and will be referred to as the "reference example". The model and calculations reproduced with THE PROGRAM will be referred to as the "verification example".

The reference example involves evaluating a floor system for vibrations induced by walking. The floor features a 16 ft cantilever; thus, it is recommended to use the Finite Element Analysis (FEA) methodology. The floor is part of a steel-framed building, and its structural system comprises a concrete slab on a steel deck supported by steel joists and girders. The floor configuration is presented in Figure 1, where the bay to be evaluated is highlighted. Two points of interest are studied. The middle span of the bay between Gridlines 3-4 and D-E, referred to as the Backspan Point; and a point close to the cantilever's edge, at the middle span between Gridlines 3-4, referred to as the Cantilever Point.

The FEA model for the verification example is created in the program according to the recommendations presented in section 7.2 of the AISC DG11. Some of these recommendations include:

- Modeling only a portion of the floor containing the bay(s) being evaluated.
- Treating beams, joists, and girders connected to concrete slabs as fully composite, meaning they work together to resist loads. Thus, effective transformed moments of inertia are used.
- Considering beam and girder shear connections as continuous.
- Accounting for orthotropic material behavior for the slabs to consider their proportional flexural stiffness in each direction. For example, this includes adding the additional stiffness provided by the steel-deck's ribs in the strong direction.

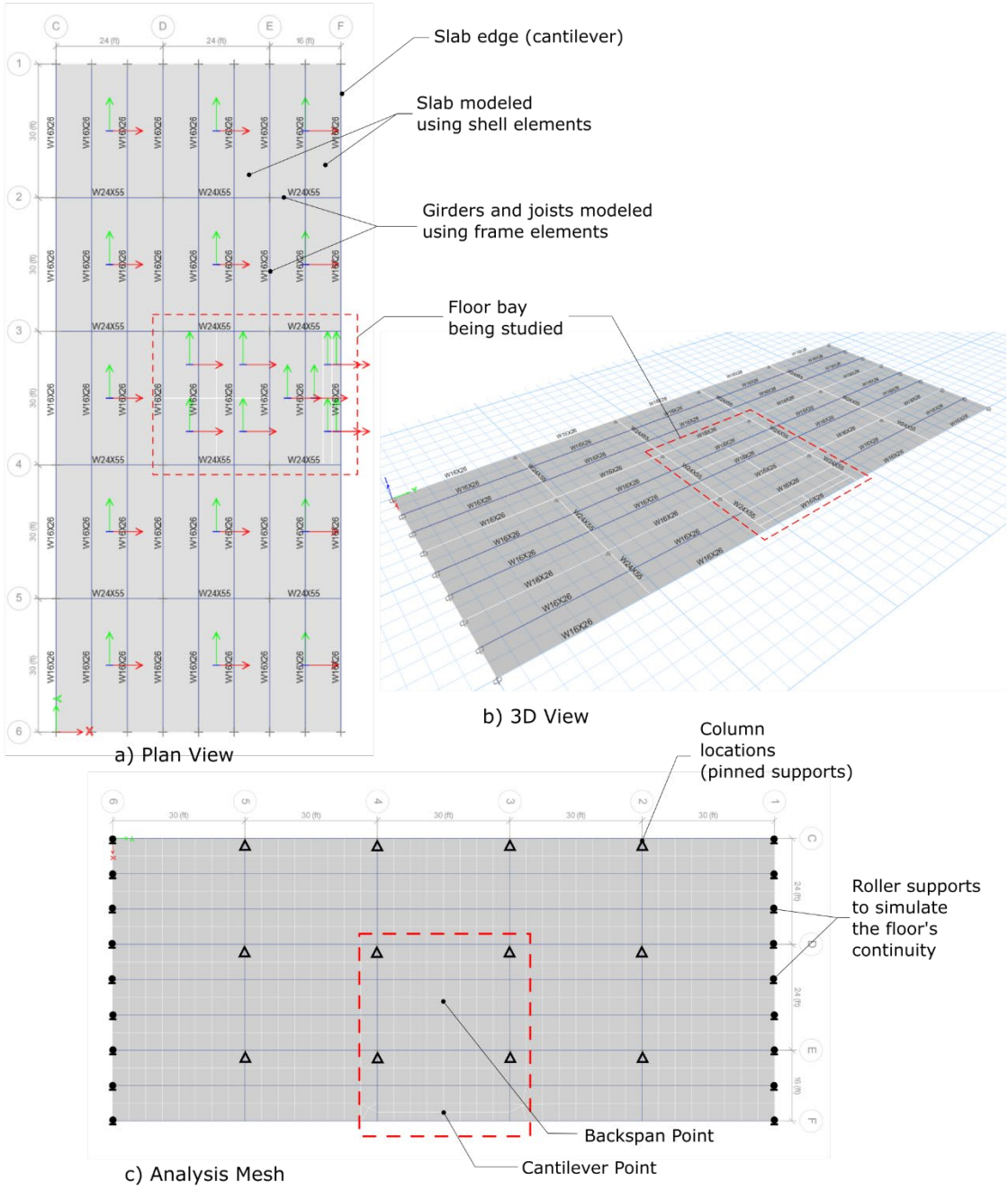


Figure 1. FEA model of the floor being evaluated for vibrations.

FEA MODEL, PROPERTIES, AND LOADING

BOUNDARY CONDITIONS

Simple (or pinned) supports are placed at the columns locations to simulate the actual support conditions. Furthermore, as the model represents only a portion of the entire floor area, roller supports are introduced at nodes on Gridlines 6 and 1, mimicking the floor's continuity. To similarly reflect continuity, the stiffness of joist elements at Gridline C is augmented by a factor of 10.

SLAB PROPERTIES

The total slab thickness is 5 ¼ in. using a 2 in. deck. Thus, the concrete slab is modeled as a thin shell element with a thickness of 3.25 in. To account for the steel-deck's ribs, the flexural stiffness in the strong direction is increased using property modifiers. In this case, the *Bending m11 modifier* of the shell section is set to 2.97. This value is obtained from the properties provided in the reference example.

The slab's material is an elastic isotropic material with modulus of elasticity E = 1.35*E_c = 2700 ksi, Poisson ratio v = 0.2, and shear modulus G = 1125 ksi.

SLAB LOADS

The slab bears a uniform load of 53 psf. This value includes its own self weight, the vibration live load corresponding to office space (per DG section 3.3), and additional dead load from ceiling, plumbing, mechanical, and electrical fixtures. This translates into a total mass of 1.65 psf-s^2/ft. For modeling simplicity, the slab's material self-weight is neglected, and the total mass is added as "area mass" to the corresponding shell elements.

JOIST AND GIRDER PROPERTIES

The joists and girders are modeled using frame elements with typical wide-flange cross-sections, their properties are imported from the AISC15 program database. Girders utilize section W24x55, while joists use section W16x26. Their properties are the following:

	W16x26	**W24x55**
Cross-Section Area [in^2]	7.68	16.2
Moment of Inertia about 3 axis [in^4]	301	1350
Moment of Inertia about 2 axis [in^4]	9.59	29.1

To account for the composite action, the stiffness of the joists and girders is increased accordingly. For the interior girders between Gridlines C and E, the moment of inertia about axis 3 (I3), is increased by a factor of 2.94. For the girders between Gridlines E and F, I3 is increased by a factor of 2.72.

Similarly, for all the joist elements (W16x26 sections), the I3 factor is set to 3.65, except those joists at the edges. Joists on Gridline C have an I3 factor of 10 for continuity purposes. Joists on the cantilever edge, at Gridline F, have an I3 factor of 8.39 that accounts for the composite action and an additional restraint due to cladding.

The factors used to increase the I3 of the beam elements are derived from the transformed moment of inertia values given in the reference example.

The material for all steel members is the standard A99Fy50 with modulus of elasticity E= 29000 ksi, Poisson ratio ν=0.3, shear modulus G = 11153.85 ksi, and density of 490 lb. / ft^3.

JOIST AND GIRDER LOADS

The self-weight of the joists and girders is automatically calculated by the program from the cross-sectional area and material density. In addition to this, a cladding-induced load of 280 lb./ft is applied at the cantilever's edge. This is represented as a "line mass" of 8.70 plf-s^2/ft, added to the joist elements on Gridline F.

VERTICAL MASS

It is important to verify that the vertical mass is properly considered in the modal analysis for the assessment of floor vibrations. The inclusion of the vertical mass can be modified by simply checking the box "include vertical mass" at the corresponding mass source definition located at the menu: **Define > Mass Source.**

MESHING

The meshing strategy ensures that interior bays measuring 24x30 ft are meshed into 6x8 elements, while cantilever bays of 16x30 ft are segmented into 4x8 elements. The layout of the resulting mesh is shown in Figure 1 (c).

For the analysis of floor vibrations, it is important to include in the mesh the point where the excitation originated and the point where the vibration's impact needs to be checked. In this example, two points of interest are required. The Backspan Point and the Cantilever Point, both illustrated in Figure 1 (c).

To generate the Backspan Point, the bay between Gridlines 3-4 and D-E is subdivided into 4 shells. Similarly, for the Cantilever Point, another division is made between Gridlines 3-4 close to the edge of Gridline F. The meshing settings in these divided shells are set to ensure uniform mesh size across all bays.

WALKING VIBRATION LOADING

Walking vibration loads are introduced through the definition of Excitation Sets (accessed through the menu: **Define > Floor Vibration > Excitation Sets**). Their definition requires specifying the Excitation Joint, the Response Joint, and other parameters related to the steady-state load case.

The parameters utilized in this verification example are shown on Figure 2, and summarize as follows.

- **Associated Joints:** The Excitation and Response Joint labels. In this case, the Backspan Joint and the Cantilever Joint were automatically labeled by the program as "56" and "66", respectively.
- **Function:** A steady-state function with a uniform magnitude of 1. The program's default "UnifSS" function serves this purpose.
- **Modal Case:** The default "MODAL" case is used, setting the maximum number of modes to 40 to capture modes up to frequencies of 20 Hz.
- **Damping:** The Stiffness proportional coefficient applied to the hysteretic damping in the steady-state load case (or the modal damping). The references example adopts a viscous damping ratio of 2.5%. This is equivalent to 5% modal damping according to the following relation:

$$d_k(\omega) = 2\zeta$$

where $d_k(\omega)$ is the stiffness proportional coefficient, and ζ is the viscous damping ratio.

- **Last Frequency:** This value represents the highest frequency used to run the steady-state analysis. A value of 20 Hz is utilized as frequencies higher than 20 Hz are unlikely to cause human discomfort due to floor vibrations.

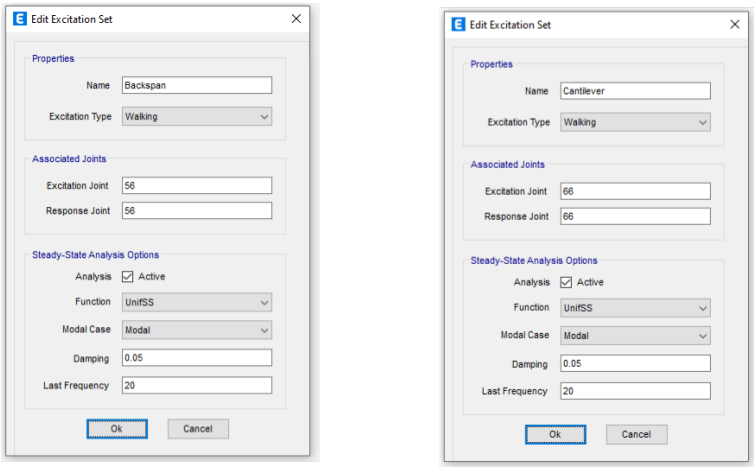


Figure 2. Definition of the Excitation Sets.

NOTES ON MODELING

When creating the FEA model, following the recommendations presented in section 7.2 of the AISC DG11 is crucial for the proper evaluation of floor vibrations. However, there are different ways of modeling the floor that will result in equivalent or similar structural properties and still be in alignment with the AISC DG11 recommendations. For instance, for steel-decks, instead of using property modifiers to account for the composite action, another alternative is to use orthotropic materials for the shell elements and setting the beam insertion point (or offsets) according to its position relative to the concrete slab.

Therefore, given the variety of modeling styles, engineering judgment should be used when selecting the most suitable strategy for each project. In this example, the modeling strategy aims to replicate the results of the reference example as close as possible.

ANALYSIS

The analysis is run after the Excitations Sets have been defined. The program automatically generates the required load cases and applies the necessary load patterns based on the definition of the Excitation Sets. The program-generated load cases for this particular example are shown in Figure 3.

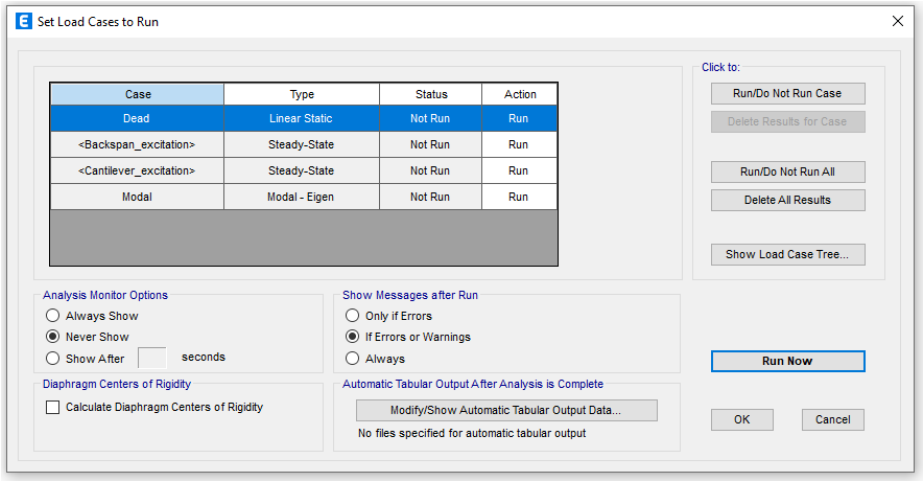


Figure 3. Program generated load cases.

TECHNICAL FEATURES TESTED

- Serviceability evaluation of a steel-framed floor subjected to vibrations induced due to walking activities according to the methodologies presented in the AISC DG 11, Chapter 7.

RESULTS COMPARISON

FEA MODEL

The first step is to compare the FEA model created for the verification example to the model that is presented in the reference example. This is done by looking at the modal frequencies. Particularly, those of modes 1,3,5,10, and 12, that are some of the most relevant modes to this example. Their relevance is deduced by noticing that the maximum vertical displacement appears to be close to either the Backspan Point or the Cantiliver Point, as seen in Figure 4.

Table 1 presents a comparison between the modal frequencies of both models, and Figure 4 visually illustrates the deformation of mode 1 and 5.

Mode	Reference Freq. (Hz)	Verification Freq. (Hz)	Difference (as %)
1	3.49	3.38	3.15%
3	4.89	4.77	2.45%
5	7.05	7.15	1.42%
10	8.85	8.57	3.16%
12	9.35	9.02	3.53%
		avg	**2.74%**

Table 1. Comparison of Modal Frequencies.

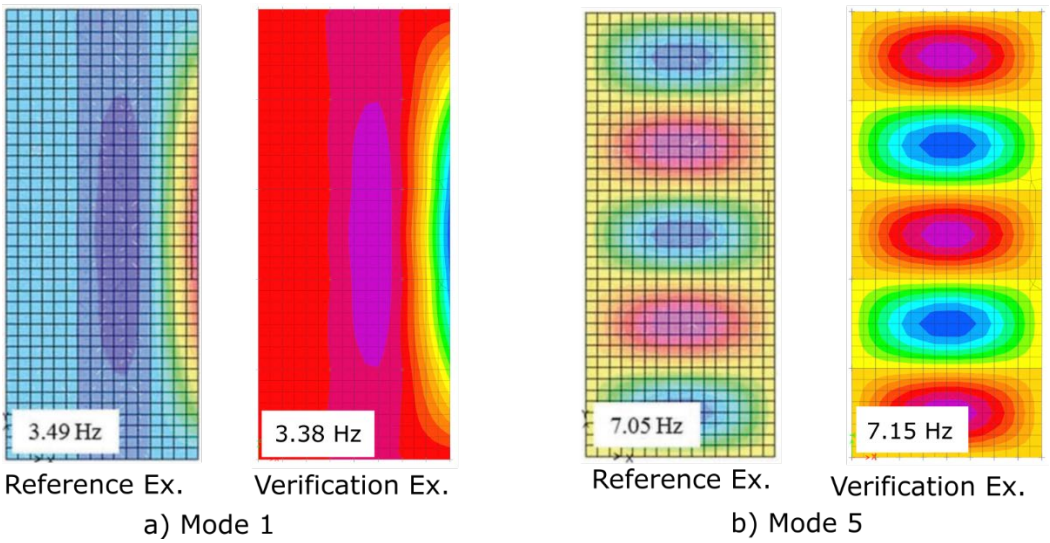


Figure 4. Visual comparison of the deformation of Modes 1 and 5. The reference example images are taken from the AISC DG11.

These comparisons reveal that both models exhibit nearly identical properties and behaviors. The observed differences are minor, with an average discrepancy of 3% in the modal frequencies. This suggests a high level of similarity between both model's dynamic responses.

Note that reproducing the analytical model with the exact same properties is difficult to achieve, as the reference example does not explicitly provide the complete set of input values and strategies utilized to develop the FEA model. As mentioned in section *Notes on Modeling*, different modeling strategies may lead to variations in the model stiffness which may have subtle impact on the modal properties.

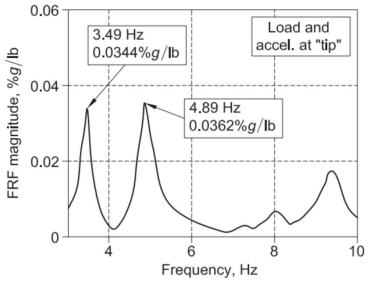
WALKING VIBRATIONS ANALYSIS RESULTS

Floors with low frequencies (below 9 Hz) primarily exhibit a resonant response, whereas floors with high frequencies (above 9 Hz) are predominantly influenced by the impulse of individual footsteps. Consequently, the analysis bifurcates into two distinct calculations to accommodate these different response characteristics. For frequencies ranging from 0 to 9 Hz, the analysis employs the Frequency Response Function (FRF) method. For frequencies between 9 to 20 Hz, it utilizes the effective impulse method. Both approaches are based in the computation of the FRF, which plots the acceleration measured at the Response Point as a result of the steady-state load pattern applied at the Excitation Point.

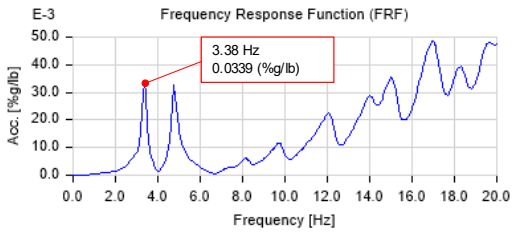
The program performs the corresponding calculations for both the low and high frequencies values of each FRF. In this verification example, the analysis focuses on the low frequency response at the Cantilever Point and the high frequency response at the Backspan Point.

CANTILEVER POINT – (LOW FREQUENCY)

The FRF for both the verification and reference examples is shown Figure 5.



a) Reference Ex.



b) Verification Ex.

Figure 5. FRF Plot for Cantilever Excitation Set. The reference example image is taken from the AISC DG11.

For the Cantilever Point, the dominant frequency f_n, at frequencies lower than 9 Hz, is located at 3.38 Hz (Mode 1) and has a magnitude of 0.0339 (%g/lb).

To compute the corresponding equivalent peak sinusoidal acceleration a_p, the following equations are used:

$$a_p = FRF_{\text{Max}} \alpha Q \rho, \qquad \textit{(AISC DG11 7-1)}$$

$$\alpha = 0.09e^{-0.075 f_n}, \qquad \textit{(AISC DG11 7-2)}$$

$$\rho = 12.5\beta + 0.625 \ \ if \ \ 0.01 \leq \beta < 0.03, \qquad \textit{(AISC DG11 7-3b)}$$

Considering a pedestrian's bodyweight of 168lb, the calculation is as follows:

$$\rho = 12.5(0.025) + 0.625 = 0.9375$$

$$\alpha = 0.09e^{-0.075(3.38)} = 0.06985$$

$$a_p = 0.0339(0.06985)(168)(0.9375) = 0.373 \ \%g$$

Verification vs Reference

From the FRF plot, it can be noted that there are two peaks below 9 Hz. The first peak corresponds to the modal frequency of mode number 1 while the second to mode number 3. In the reference example, the second peak is slightly higher than the first one. On the other hand, in the verification example, the first peak is the one that is slightly higher (see Figure 5).

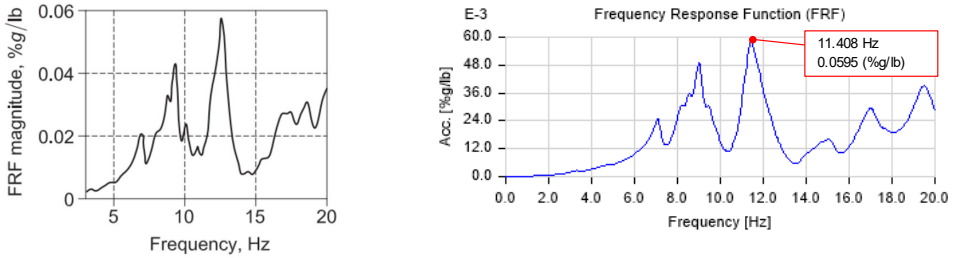
According to the AISC DG11, the dominant frequency is taken as the frequency with the maximum FRF value. Therefore, for the reference example the dominant frequency is 4.89 Hz, and for the verification it is 3.39 Hz. However, the reference example calculates the predicted sinusoidal acceleration a_p for both peaks. Thus, a direct comparison of the results at the first peak (corresponding to the frequency of mode number 1) is possible. Such comparison is provided in Table 2. The acceleration tolerance limits are taken from the values presented in the AISC DG11, included in the Appendix as Fig A1. For this case, the tolerance limit is 0.5 %g.

	FRF at mode 1 (%g/lb.)	Predicted Acc. a_p (%g)	Tolerance Limit (%g)	DCR
Reference (AISC DG11)	0.0344	0.374	0.5	0.750
Verification (SAFE/ETABS)	0.0339	0.373	0.5	0.746

Table 2. Comparison of results for the Cantilever Point (considering low frequencies)

BACKSPAN POINT – (HIGH FREQUENCY)

The FRF for both the verification and reference examples is shown Figure 6. For the Backspan Point, the dominant frequency f_n, at frequencies higher than 9 Hz, is located at 11.408 Hz (Mode 19) and has a magnitude of 0.0595 (%g/lb).



a) Reference Ex.

b) Verification Ex.

Figure 6. FRF Plot for the Backspan Excitation Set. The reference example image is taken from the AISC DG11.

To compute the equivalent peak sinusoidal acceleration a_{ESPA}, the following equations are used:

$$a_{p,m} = 2\pi f_{n,m} \Phi_{i,m} \Phi_{j,m} I_{\text{eff},m} \qquad \textit{(AISC DG11 7-4)}$$

$$I_{eff,\,m} = \left(\frac{f_{\text{step}}{}^{1.43}}{f_{n,m}{}^{1.30}}\right)\left(\frac{Q}{17.8}\right) \qquad \textit{(AISC DG11 1-6)}$$

$$f_{step} = f_n / h$$

$$a(t) = \sum_{m=1}^{N_{\text{modes}}} a_{p,m}\, e^{-2\pi f_{n,m}\beta t}\, sin\left(2\pi f_{n,m} t\right) \qquad \textit{(AISC DG11 7-5)}$$

$$a_{\text{ESPA}} = \sqrt{2\left[\frac{1}{T}\int_0^T [a(t)]^2 dt\right]} \approx \sqrt{2}\sqrt{\frac{1}{N}\sum_{k=1}^{N} a_k^2} \qquad \textit{(AISC DG11 7-6)}$$

$$\Delta t = 0.005$$

At each k-th step, $a_k = a(\Delta t \cdot k).$

$$T = 1/f_{step},$$

The calculations involve a summation over the discretized period *T (Eq. AISC DG11 7-6)*. This equation involves a second nested summation over all the modes

covering frequencies up to 20 Hz *(Eq. AISCD G11 7-5)*. To perform these calculations, the displacements in the vertical (GLOBAL-Z) direction for both the Excitation and Response Points are required ($\Phi_{i,m}\Phi_{j,m}$). This information is provided in the Appendix at Table A1.

The harmonic value h, required to compute the step frequency f_{step}, is taken as h=6. This value is linked to the dominant frequency, and it is obtained from Table 7-1 in the AISC DG11, included here in the Appendix as Table A2.

The pedestrian bodyweight is taken as 168 lb. as recommended in the AISC DG11.

Putting all these values into equation AISC DG11 1-6, 7-4, 7-5, and 7-6, the equivalent peak sinusoidal acceleration is computed as:

$$a_{\text{ESPA}} = 0.296\ \%\text{g}$$

Verification Example vs Reference Example

From the FRF, the dominant frequency is identified to match the frequency of mode number 22 for the reference example, and mode 19 for the verification example. Despite being a different mode number and having a slightly different modal frequency, both models are consistent, showing that the max FRF peak corresponds to the same modal shape in each model, as it is shown in Figure 7.

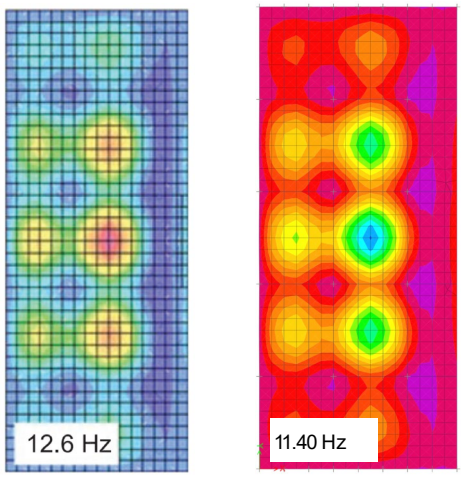


Figure 7. Visual comparison of the deformation of Mode 22 in the reference example to Mode 19 in the verification example. The reference example image is taken from the AISC DG11.

The acceleration tolerance limits are taken from the values presented in the AISC DG11, included in the Appendix as Fig A1. For this case, the tolerance limit is interpolated according to the dominant frequency, which is slightly different for each model. The results are summarized in the Table 3.

	FRF max (%g/lb.)	Predicted Acc. a_{ESPA} (%g)	Tolerance Limit (%g)	DCR
Reference (AISC DG11)	≈0.057	0.314	0.78	0.40
Verification (SAFE/ETABS)	0.0595	0.296	0.71	0.42

Table 3. Comparison of results for the Backspan Point (considering high frequencies)

COMPUTER FILE

AISC-DG11 2ED Ex001

CONCLUSION

For the low frequency analysis, based on the FRF method, the results are almost identical with a DCR of 0.75 in both examples. Here it is important to point out that there is a minor difference in the maximum FRF value which dictates the dominant frequency. The reason is that the maximum FRF value seems to be repeated at two different frequencies, corresponding to modes number 1 and 3. However, the comparison of results is carried on at the same frequency value, corresponding to mode number 1. Thus, the comparison is valid.

In the high frequency analysis, which employs the effective impulse method, a notable observation is the shift in the dominant frequency to a different mode number between the two models. Specifically, the model used in the reference example identifies the dominant frequency at Mode 22, with a frequency of 12.6 Hz, whereas the verification example locates it at Mode 19 with a frequency of 11.40 Hz. A closer inspection reveals that these modes refer to the same modal shape (see Figure 7). Thus, validating the comparison. Furthermore, despite the difference in the mode frequency, both the reference and the verification example arrive at a similar DCR value of 0.40, and 0.42, respectively.

In conclusion, the obtained results in the program show good agreement with example 7.1 presented in the AISC DG11. The minor differences in the results are likely attributable to the variations in the modal properties that are result of employing different modeling strategies when constructing the FEA model.

APPENDIX

Table A1. Modal information for the Backspan Excitation Set. The field "Modal Uz" corresponds to the vertical displacement of either the Excitation or Response Joint at the given *nth* mode.

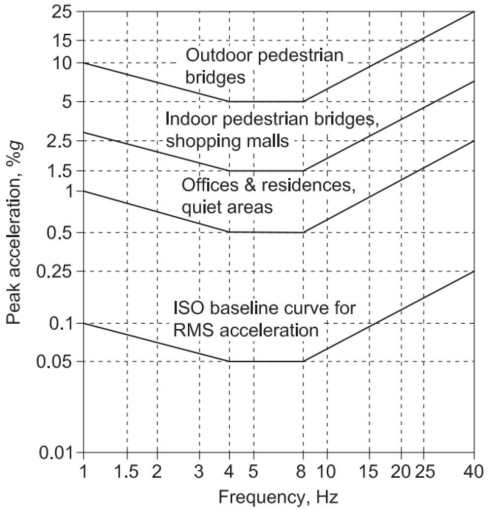
Mode	Frequency (Hz)	Period (s)	Modal Uz (Excitation)	Modal Uz (Response)
1	3.382	0.296	-0.013864	-0.013864
2	3.701	0.27	0.000001	0.000001
3	4.771	0.21	-0.015741	-0.015741
4	6.575	0.152	0.000008	0.000008
5	7.154	0.14	-0.054115	-0.054115
6	7.435	0.134	0.021603	0.021603
7	7.695	0.13	0.000009	0.000009
8	7.858	0.127	0.000012	0.000012
9	8.217	0.122	-0.045528	-0.045528
10	8.571	0.117	0.054949	0.054949
11	8.898	0.112	0.000039	0.000039
12	9.021	0.111	-0.086888	-0.086888
13	9.131	0.11	3.17E-07	3.17E-07
14	9.482	0.105	-0.056053	-0.056053
15	9.811	0.102	0.033772	0.033772
16	10.23	0.098	0.021617	0.021617
17	10.438	0.096	-0.000007	-0.000007
18	10.505	0.095	0.000004	0.000004
19	11.408	0.088	0.097315	0.097315
20	11.762	0.085	-0.058761	-0.058761
21	12.056	0.083	0.000108	0.000108
22	12.177	0.082	0.032975	0.032975
23	12.812	0.078	-0.000023	-0.000023
24	13.225	0.076	0.000014	0.000014
25	13.75	0.073	0.008611	0.008611
26	14.099	0.071	-0.02598	-0.02598
27	14.636	0.068	-0.032083	-0.032083
28	15.111	0.066	0.0426	0.0426
29	15.49	0.065	0.000083	0.000083
30	15.778	0.063	-0.000068	-0.000068
31	16.333	0.061	-0.000006	-0.000006

32	16.697	0.06	0.015102	0.015102
33	17.09	0.059	0.058743	0.058743
34	17.746	0.056	-0.03418	-0.03418
35	18.337	0.055	0.022798	0.022798
36	18.48	0.054	-0.000513	-0.000513
37	19.104	0.052	-0.000233	-0.000233
38	19.637	0.051	0.068291	0.068291
39	19.947	0.05	0.000053	0.000053
40	20.356	0.049	-0.000047	-0.000047

Table A2. Harmonic Selection for High-frequency Floors. Based on Table 7-2 from the AISC DG11.

Dominant Frequency, Hz	***h***
9-11	5
11-13.2	6
13.2-15.4	7
15.4-17.6	8
17.6-20	9

Fig A1. Recommended Tolerance limits for human comfort. Taken from Fig 2-1 in the AISC DG11.



BIBLIOGRAPHY

Murray, Thomas M., David E. Allen, Eric E. Ungar, and D. Brad Davis. 2016, AISC Design Guide 11 Vibrations of Steel-Framed Structural Systems Due to Human Activity. 2nd ed. Chicago, Illinois: American Institute of Steel Construction.